

MOBILE BROOKLEY, AL, USA

WMO: 722235

Lat: 30.626N

Lon: 88.068W

Elev: 26

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13838

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.1	33.2	14.2	11.3	36.7	18.7	14.1	40.1	24.0	43.3	20.9	47.4	7.5	330	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.1	93.8	78.4	92.0	78.0	90.5	77.7	81.5	88.7	80.6	87.6	79.9	86.9	7.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.5	153.0	84.7	78.9	150.3	84.4	77.9	145.3	83.8	45.1	89.1	44.1	87.8	43.2	86.5	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.2	17.8	16.0	23.2	97.0	3.6	2.6	20.6	98.8	18.5	100.3	16.5	101.8	13.9	103.7	
			DB	20.7	83.1	3.2	1.2	18.4	84.0	16.5	84.7	14.7	85.4	12.3	86.3
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.3	51.2	54.9	61.0	67.4	74.8	80.6	82.4	82.5	79.3	70.2	60.0	53.9
	DBStd	13.01	9.32	8.38	7.63	5.83	4.85	3.06	2.55	2.81	4.18	7.25	7.81	9.00
	HDD50	227	100	42	10	0	0	0	0	0	0	0	12	63
	HDD65	1517	433	293	167	43	3	0	0	0	0	35	187	356
	CDD50	6893	137	180	350	522	768	919	1005	1009	880	627	312	184
	CDD65	2708	5	11	42	114	306	469	540	544	430	198	37	12
	CDH74	25584	4	21	137	554	2515	4845	5847	5900	4126	1473	146	16
	CDH80	9288	0	0	10	49	666	1881	2419	2457	1457	338	11	0
(14) Wind	WSAvg	7.1	7.8	7.9	8.1	8.1	7.1	6.0	5.3	5.6	7.0	7.4	7.4	7.5
(15) Precipitation	PrecAvg	65.8	5.5	4.5	6.1	5.4	5.0	6.0	8.2	6.5	5.4	3.3	4.5	5.2
	PrecMax	84.7	16.1	10.2	15.7	17.7	14.9	17.8	22.7	12.8	21.7	9.5	11.7	14.5
	PrecMin	42.2	1.0	0.7	0.6	0.6	0.4	1.7	3.9	1.1	2.0	0.0	1.3	1.3
	PrecStd	11.6	3.6	2.4	3.4	4.1	3.2	3.5	3.7	3.0	4.2	2.6	2.9	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.9	77.1	81.3	83.9	91.5	95.2	96.3	96.4	94.8	89.1	81.4	76.2
		MCWB	66.8	67.6	66.5	70.6	73.5	77.0	79.2	78.3	77.8	75.4	72.9	69.1
	2%	DB	70.5	72.8	77.9	81.4	88.1	92.4	93.2	93.5	91.0	86.1	78.4	73.1
		MCWB	65.4	66.8	67.2	69.8	73.7	77.3	78.9	79.0	76.6	74.0	70.9	68.2
	5%	DB	68.0	70.2	74.7	79.4	85.7	90.4	91.3	91.5	89.5	83.6	75.0	70.7
		MCWB	64.3	65.0	66.3	69.3	73.8	77.1	78.3	78.7	76.2	72.8	68.1	67.0
	10%	DB	65.5	67.5	72.4	77.3	83.7	88.5	89.9	90.1	87.6	81.4	72.6	67.8
		MCWB	62.5	63.3	65.8	68.7	73.5	76.7	78.3	78.4	75.9	72.2	66.2	64.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.5	71.2	72.3	76.1	79.2	81.7	82.5	83.1	81.9	79.8	76.1	72.2
		MCDB	71.7	73.5	76.3	79.7	84.2	89.2	90.4	90.9	87.8	83.6	79.2	74.2
	2%	WB	67.2	68.9	70.6	74.3	77.6	80.4	81.3	81.7	80.3	77.9	72.8	70.0
		MCDB	69.3	71.4	74.1	78.4	83.7	87.6	88.7	89.0	86.2	82.8	76.2	71.9
	5%	WB	65.3	66.6	69.1	72.8	76.4	79.5	80.3	80.6	79.0	76.2	70.4	67.5
		MCDB	67.1	68.8	72.7	76.8	82.5	86.4	87.8	87.8	85.2	81.3	73.7	70.0
	10%	WB	63.1	64.5	67.2	71.3	75.3	78.4	79.5	79.8	78.0	74.7	68.3	65.2
		MCDB	64.8	66.5	70.7	75.1	81.4	85.2	86.9	86.9	84.2	79.5	71.5	67.3
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	17.7	18.3	17.5	16.8	15.2	15.0	15.1	15.8	18.7	19.6	17.5
		MCDBR	17.6	17.2	18.3	17.8	18.2	17.9	17.0	17.1	17.3	18.1	19.0	17.2
		MCWBR	13.2	11.8	10.5	9.5	7.6	6.9	6.6	6.7	6.8	8.5	11.9	13.3
	5% WB	MCDBR	14.8	14.4	14.4	13.7	14.2	15.0	15.1	15.0	14.1	13.7	15.6	15.6
		MCWBR	12.6	11.9	10.1	9.1	7.1	6.9	6.9	6.9	6.4	7.7	12.0	13.4
(40) Clear-Sky Solar Irradiance	taub		0.331	0.349	0.373	0.403	0.439	0.465	0.514	0.503	0.447	0.383	0.355	0.340
	taud		2.547	2.500	2.427	2.346	2.280	2.258	2.140	2.167	2.291	2.454	2.476	2.531
	Ebn at Noon		286	290	289	283	272	263	250	251	262	274	274	276
	Edh at Noon		27	31	36	40	43	44	49	47	40	32	29	26
(44) All-Sky Solar Radiation	RadAvg		963	1141	1543	1880	2078	1987	1876	1764	1650	1440	1100	846
	RadStd		75	157	123	150	111	151	128	126	102	148	96	76

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+506	+324
(47) Regional (31 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+263	+167

Nomenclature: See separate page